Cross-Discourse Authorship Verification under Human–AI and Imitative-AI Conditions

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Abstract

Traditionally, Authorship Verification (AV) assumes that every writer has a unique and stable stylistic fingerprint. However, the rise of Large Language Models (LLMs) as "style translators" creates a new challenge for forensic linguistics. This study explores how AV systems perform when facing deliberate style imitation. We built a progressive evaluation framework using the HC3 Plus, MixSet, and MULTITuDE datasets (Su et al., 2024; Zhang et al., 2024; Macko et al., 2023). Our experiments cover three scenarios: Human-Human, Human-AI (Free Generation), and Human-AI (Explicit Imitation). We used TF-IDF character n-grams and word-level features to measure performance drops across semantic, structural, and physical attacks.

Our results show a "staircase-like" decay in detection accuracy. Native AI text is easy to identify, with an AUC of 0.996. However, when GPT-4 performs explicit token-level imitation, performance drops to 0.678, which is near-random guessing. We also found that semantic-level paraphrasing is the most dangerous strategy. It leads to a 17.6% drop in AUC. These findings prove that statistical fingerprints are vulnerable in the post-LLM era. We suggest that future AV systems should focus on higher-order features, such as discourse structure and logical consistency.

Keywords: Authorship Verification, Large Language Models, Adversarial Attacks, Stylometry, MULTITuDE.

1 Introduction

In recent years, large language models (LLMs) have made significant progress in generating natural language (Su et al., 2024). From open-ended question answering and summarization to creative writing, texts produced by these models can be as fluent and readable as those written by humans. This improves content production efficiency, but also poses new challenges to computational forensics and authorship verification (AV) (Bevendorff et al., 2024; Macko et al., 2024).

Traditional authorship verification methods assume that human writers leave stable stylistic fingerprints when writing naturally and unconsciously (Stamatatos, 2009). These fingerprints appear in features such as character distributions, word choices, function word frequency, and syntactic preferences. Based on this idea, Stamatatos (2009) points out that character n-gram features are robust for author identification across topics and genres. Therefore, such methods are widely used in forensics, plagiarism detection, and anonymous text attribution (Bevendorff et al., 2019).

However, these methods assume that authors write without external intervention. Modern LLMs challenge this assumption. Unlike early template- or rule-based text generators, LLMs can generate coherent text and also control style or perform style transfer (Zhang et al., 2024; Su et al., 2024). With fine-tuned instructions or contextual examples, a model can imitate the style of a specific author, domain, or community. In this sense, LLMs act as a "style translator": they rewrite surface stylistic features while keeping the meaning largely unchanged (Zhang et al., 2024).

This style obfuscation threatens current author identification systems (Emmery et al., 2018). If LLMs align human-style features at character or word level, detectors that rely on statistical features may fail (Zhang et al., 2024). Beyond semantic rewriting, attackers can also use back-translation or physical-level perturbations, such as homograph attacks, to hide the text source further.

This leads to a key question: how robust are current author identification and AI detection methods against deliberate, layered style imitation and multidimensional attacks?

To answer this, this paper studies the effects of generation mode, modification granularity, and adversarial attacks on the robustness of author identification systems. The goal is not only to distinguish human text from AI text, but to examine how forensic clues in text are weakened under adversarial conditions. We use the MULTITuDE adversarial framework (Macko et al., 2024) to test detectors on semantic, structural, and physical attacks.

This paper addresses three research questions:

- **Question1**: How do different generation modes (free generation, semantically unchanged generation, explicit style imitation) affect the original stylistic features of text?
- **Question2**: How does modification granularity (sentence-level vs. lexical/character-level) affect the performance of statistical author identification and AI detection models?
- **Question3**: What are the robustness limits of existing detectors under semantic, structural, and physical attacks? How can the threat levels of different attacks be ranked?

We conduct systematic experiments using HC3 Plus (Su et al., 2024), MixSet (Zhang et al., 2024), and MULTITuDE adversarial datasets. Our goal is to provide a realistic threat model and analysis for author identification in the post-LLM era. To ensure reproducibility and support further forensic research, I have put the evaluation pipeline and experimental scripts on `https://github.com/Joyoem/Human_AI_Authorship_Verification`.[1]

2 Related Work

2.1 Traditional Stylometry and Authorship Verification

Authorship verification (AV) is a core topic in computational and forensic linguistics. It is based on the idea that human authors leave stable and repeatable stylistic features in their writing, even when writing naturally and without thinking (Stamatatos, 2009). These features are shaped by long-term habits, not conscious choices, so they can be useful for forensic analysis

Stamatatos (2009) summarized the methods in this field and pointed out that character n-grams are especially robust. They can capture spelling habits, use of affixes, and local syntax patterns. This makes them strong indicators for authorship across different topics and genres. Koppel et al. (2009) clarified the difference between authorship attribution and authorship verification, establishing a clear framework for research in this area.

Before generative AI became widespread, AV methods based on stylistic features were considered the "gold standard" and were widely used in forensic investigations.They were widely used in forensic investigations, plagiarism detection, and anonymous text analysis (Bevendorff et al., 2019). Their reliability came from the stability and interpretability of stylistic features in normal, non-adversarial texts.

2.2 Machine-Generated Text Detection

With the rise of large language models (LLMs), detecting machine-generated text (MGT) has become an important research topic (Zhao et al., 2023). Early methods relied on differences in statistical distributions. For example, GLTR (Gehrmann et al., 2019) visualized word prediction ranks to find abnormal patterns. DetectGPT (Mitchell et al., 2023) used a zero-shot method based on probability curvature to identify generated text.

These methods work well in simple scenarios but struggle in more complex or adversarial settings. MGTBench (He et al., 2024) showed that detectors often fail when applied to new models or settings. Macko et al. (2023) proposed the MULTITuDE multilingual benchmark, which added adversarial perturbations to test detectors. They found that paraphrasing or translation can hide the statistical patterns that detectors rely on (Macko et al., 2023).

Although MGT detection is not the same as author identification, its findings are useful for AV research. Text that has been systematically rewritten can hide statistical signals. LIFE (Wang et al., 2025) tries to extract deeper model fingerprints using prompts, but its effectiveness in real-world adversarial conditions is still under study.

2.3 Adversarial Obfuscation and Stylistic Imitation

Adversarial obfuscation aims to hide or weaken the stylistic features used by author identification systems(Emmery et al., 2018). Early approaches were rule-based or heuristic (Bevendorff et al., 2019). With LLMs, this has evolved into more advanced style transfer and imitation.

Attacks are usually divided into three types: **semantic rewriting** changing wording without al-

[1] `https://github.com/Joyoem/Human_AI_Authorship_Verification`

tering meaning (e.g., LLM rewriting), **structural transformation** changing sentence or paragraph structure (e.g., back-translation), and **physical character changes** changing letters without changing meaning (e.g., homographs) (Macko et al., 2024).

PAN 2024 introduced the Voight–Kampff task (Bevendorff et al., 2024) to test these attacks in AV evaluation. This marks a shift from static classification to dynamic adversarial scenarios.

Even with defenses like meta-contrastive learning (Lv et al., 2024) and R-Drop regularization (Lin et al., 2024), risks remain. MixSet (Zhang et al., 2024) shows that fine-grained token-level imitation can erase human text statistical features. Building on this, our study combines HC3 Plus semantic imitation (Su et al., 2024) with the three-dimensional MULTITuDE framework to analyze how different attack types and modification levels affect statistical feature detectors.

2.4 Statistical vs. Neural Obfuscation

Current literature distinguishes between statistical and neural approaches to authorship verification. While neural models like BERT excel at semantic understanding, they are highly vulnerable to adversarial "style washing" (Zhao et al., 2023). Stamatatos (2009) maintains that statistical features (n-grams) remain the most robust forensic evidence because they target granular writing habits that LLMs often overlook during high-level rewriting.

Furthermore, as noted in the MULTITuDE benchmark, the challenge of detecting machine-generated text increases across different languages. In morphologically rich languages, word-level detectors often fail, making character-based n-grams a critical tool for cross-linguistic forensic analysis (Macko et al., 2023). Our work builds on this by testing how explicit imitation strategies specifically degrade these low-level statistical markers.

3 Methodology

3.1 Dataset Selection and Strategy

We gradually built our dataset to support both benchmark comparison and targeted adversarial evaluation (Macko et al., 2023). To avoid relying only on simple prompt-based imitation, we combined three specialized datasets (Su et al., 2024; Zhang et al., 2024):

1. **Style Imitation Datasets**:
   - **HC3 Plus** Su et al. (2024) provides a semantic-invariant parallel corpus,. We use it as the core dataset for style imitation.
   - **MixSet** Zhang et al. (2024) contains sentence-level and token-level explicit style attack samples. These allow us to analyze how different modification influence stylistic features.
2. **Adversarial Robustness Dataset**:
   - **MULTITuDE** Macko et al. (2023) is used to test detectors under complex adversarial conditions. It includes systematically obfuscated parallel texts designed to challenge the detector on three dimensions: semantics, structure, and physical character changes.

3.2 Progressive Adversarial Evaluation Framework

To simulate the increasing intensity of text "style whitening," we built a progressive evaluation framework:

3.2.1 Three Levels of Style Testing

- **Free Generation**: The model generates text without constraints. This highlights the strongest machine-generated fingerprints.
- **Semantic-Invariant Generation**: Text is generated under tasks like translation or summarization. This simulates natural dilution of stylistic features(Su et al., 2024).
- **Explicit Imitation**: Fine-grained word or token-level replacements imitate a target style. This tests detector performance under strong adversarial conditions. (Zhang et al., 2024).

3.2.2 Robustness Stress Test

Using the MULTITuDE framework, we compare detection performance on original vs. obfuscated texts (Macko et al., 2023).

As shown in Figure 1, this framework quantifies how different strategies, such as paraphrasing or back-translation, systematically erase stylistic fingerprints. By testing across multiple attack categories, we can identify which "style translation" methods pose the greatest threat to stylometric stability (Macko et al., 2024).

Figure 1: The MULTITuDE adversarial framework used for robustness evaluation, illustrating the transformation from original human/AI text to various obfuscated versions (Macko et al., 2023).

3.3 Evaluation Pipeline

We design an evaluation process that balances interpretability and reliability (Stamatatos, 2009).

3.3.1 The Forensic Significance of Sub-word Features

The choice of character-level features over word-level features is a strategic decision in forensic linguistics. While word-level choice can be consciously manipulated by a writer or an LLM, sub-word character sequences often reflect deeply ingrained, non-conscious habits (Stamatatos, 2009).

For instance, the use of specific 3-gram prefixes (e.g., `pre-`, `un-`) or 5-gram suffixes (e.g., `-ingly`, `-ation`) is often tied to an individual's level of education and linguistic background (Grieve, 2007). In the context of LLM attacks, GPT-4 is capable of high-level semantic rewriting, but it often standardizes these micro-level character distributions according to its training corpus's median frequency. By analyzing the TF-IDF weight shifts of these specific n-gram clusters, we can pinpoint exactly where the machine's "style translator" begins to overwrite the human's unique structural markers (Stamatatos, 2009; Bevendorff et al., 2020).

3.3.2 Feature Extraction

We extract two complementary feature sets:

- **Character-based features**: 3–5 order character n-grams with TF–IDF weighting. These capture micro-level spelling patterns and local syntactic preferences(Stamatatos, 2009).
- **Word-based features**: 1–2 order word n-grams. These reflect higher-level word choice and collocation preferences, especially under adversarial modifications.

3.3.3 Classifier and Metrics

We use **Logistic Regression** for its interpretable weights. Performance is measured using **ROC-AUC** and the **Brier Score** to evaluate prediction confidence.

3.4 Mathematical Formulation

3.4.1 Mathematical Rationale of Stylometric Features

To quantify the stylistic differences between human and machine text, we employ a TF-IDF weighted vector space model. According to Stamatatos (2009), character-level n-grams are particularly effective because they capture sub-word linguistic habits that are often unconscious.

We specifically selected character n-grams of order 3 to 5. Tri-grams (order 3) help capture root syllables, while penta-grams (order 5) are sensitive to morpho-syntactic markers, such as specific prefixes, suffixes, and punctuation usage patterns (Stamatatos, 2009). The weight for each term t in document d is defined as follows:

$$Weight(t,d) = TF(t,d) \cdot \log \frac{N}{DF(t)} \quad (1)$$

Where:

- **Term Frequency (TF)**: Represents how often a specific n-gram appears in a text. A high TF indicates a stylistic preference of the author.
- **Inverse Document Frequency (IDF)**: Acts as a filter to penalize "stylistic noise." Common sequences like "the" or "ing" receive low IDF scores, while unique markers receive higher weights (Bevendorff et al., 2020).

This mathematical approach allows the detector to ignore universal language patterns and focus on the **idiosyncratic fingerprints** that distinguish one author from another.

4 Results and Analysis

4.1 Staircase-like Decay of Detection Performance

Our experiments show a clear **stepwise decay** in detection performance as adversarial strength increases. Table 1 summarizes the performance across different generation modes.

- **Free Generation**: In unconstrained generation, the TF–IDF character n-gram detector performs almost perfectly, with an AUC of 0.9966 and a Brier Score of 0.0242. This shows that machine-generated text (MGT) and human writing still differ significantly

Scenario (Pair)	Model	AUC	Brier Score
Free Gen (Pair 1)	LLaMA-3	0.9966	0.0242
Semantic-Inv (Pair 2)	GPT-4	0.8462	0.1254
Hard Imitation (Pair 3)	LLaMA-3	0.7423	0.1831
Hard Imitation (Pair 3)	GPT-4	0.6788	0.2281

Table 1: Detection performance degradation across different generation and imitation levels based on HC3 Plus and MixSet datasets

in character distributions, confirming the effectiveness of traditional stylometry in non-adversarial settings (Stamatatos, 2009).

- **Semantic-Invariant Generation**: When semantic constraints (like translation or summarization) are applied, detection drops to an AUC of 0.8462. The model does not get explicit style instructions, but task requirements push it to use vocabulary closer to human norms. This naturally reduces extreme machine fingerprints (Su et al., 2024).

- **Explicit (Hard) Imitation**: Under fine-grained token-level style imitation, the detector struggles. For GPT-4 generated texts, the AUC drops to 0.6788, close to random guessing. This shows that LLMs with precise style control can systematically remove machine attributes in character n-gram space, weakening detectors based on statistical features (Zhang et al., 2024).

4.2 Comparative Analysis: GPT-4 vs. LLaMA

We compared the concealment capabilities of GPT-4 and LLaMA-3 under style imitation attacks.

- **Detection Accuracy**: In the "Hard" scenario, GPT-4 shows stronger concealment. Its detection accuracy drops to 0.6234, while LLaMA remains at 0.7423. This indicates that larger models follow instructions better and align with human-style distributions more effectively (Bevendorff et al., 2024).

- **Probabilistic Confidence**: Brier Score shows the "confidence collapse" phenomenon. Under GPT-4 Hard attack, the Brier Score reaches 0.2281. This means the detector often predicts incorrectly and outputs probabilities around 0.5. High-entropy predictions indicate successful style obfuscation. GPT-4 can mimic human-like probability distributions effectively (Zhang et al., 2024).

4.3 Robustness Benchmarking under the MULTITuDE Framework

We test the generality of our findings using the MULTITuDE adversarial framework (Macko et al., 2024). By applying various attack dimensions, we evaluate the limits of statistical detection models.

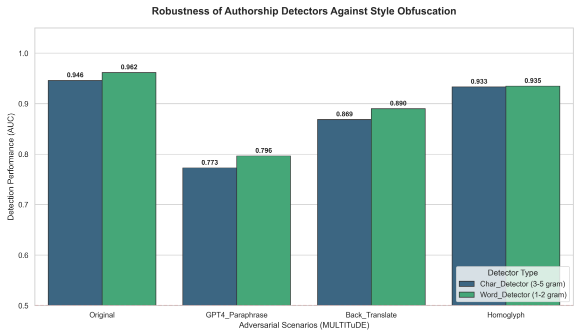

Figure 2: Robustness of authorship detectors across different adversarial scenarios. GPT-4 paraphrasing represents the most severe threat to detection stability.

As shown in Figure 2, the experimental results reveal significant differences in detector vulnerability:

- **Original (No Obfuscation)**: Both character and word-based detectors perform near perfectly, with a Word-based AUC of **0.9617** and a Char-based AUC of **0.9459**. This confirms strong statistical separability in native machine-generated text.

- **GPT-4 Paraphrase**: This semantic attack causes the largest drop in performance, with the char-based AUC falling to **0.7726**. This supports our claim that LLMs can act as "style translators" to hide their traces.

- **Structural and Physical Attacks**: **Back-translation** causes a moderate decline (AUC $\approx$ 0.8686), while **Homoglyph** attacks have almost no impact (AUC $>$ 0.93). This indicates that n-gram detectors have passive robustness against non-semantic character noise.

4.3.1 Feature-level Robustness Comparison

Our data shows that **word-level features (1-2 gram)** are more robust than character-level features in all adversarial scenarios. For example, under GPT-4 rewriting, word-based AUC = 0.7965 vs. char-based AUC = 0.7726. LLMs can rewrite local character sequences, but higher-level word patterns and function word distributions remain harder to remove.

Attack	Detector	AUC	Type
Original	Char-based	0.9459	None
Original	Word-based	0.9617	None
GPT4-Para	Char-based	0.7726	Semantic
GPT4-Para	Word-based	0.7965	Semantic
Back-Trans	Char-based	0.8686	Structural
Homoglyph	Char-based	0.9331	Physical

Table 2: Detailed performance of detectors under multi-dimensional attacks.

5 Discussion

5.1 The Impact of Modification Granularity

Our experiments show that **modification granularity** is the most important factor in style obfuscation.

- **Sentence-level rewriting**: This mainly improves fluency and semantic meaning. It does not strongly disrupt the underlying character-level statistics.

- **Token-level rewriting (Hard)**: This mode directly targets the n-gram features used by the detector. For GPT-4, this caused the AUC to drop from approximately 0.91 to 0.67.

These results support the "style-invariant feature" idea described by Emmery et al. (2018). Fine-grained control at the token level is much more effective is much more effective in hiding machine traces.

5.2 Information Entropy and Detector Confidence

Our analysis of the Brier Scores reveals a critical phenomenon: the collapse of detector confidence under adversarial pressure. In a non-adversarial setting (Pair 1), the detector output is highly polarized, with probabilities clustering near 0 or 1. This indicates low entropy and high certainty (Zhang et al., 2024).

However, during a GPT-4 "Hard" imitation attack, the probability distribution shifts toward the 0.5 mark, resulting in a Brier Score of 0.2281. This suggests that the LLM is not just creating a new style, but effectively "neutralizing" the stylistic signal until it reaches a state of maximum uncertainty for the detector. From a security standpoint, this means that the "style translator" acts as a noise generator that targets the specific frequency bands where authorship fingerprints reside (Wang et al., 2025; Zhao et al., 2023).

5.3 Efficacy of Adversarial Dimensions

Using the MULTITuDE framework, we can rank the threat levels of different attack types:

1. **Semantic Attacks**: GPT-4 Paraphrasing is the most powerful method. It rewrites expressions while keeping content identical, causing the word-level AUC to drop from 0.962 to 0.796.

2. **Structural Attacks**: Back-translation changes sentence structure but preserves semantics. Detectors still perform relatively well (AUC $\approx$ 0.89) because many word patterns remain unchanged.

3. **Physical Attacks**: Homoglyph substitutions have almost no effect on performance. This proves that statistical features are resilient against non-semantic noise.

5.4 Limits of Style Translators and Defense Implications

Even though GPT-4 can mimic styles at the character level and fool statistical detectors, it does not achieve perfect style replication.

- Higher-level features, such as word distributions and discourse patterns, can still reveal anomalies (Wang et al., 2025).

- Our results show that word-level detectors outperform character-level detectors under adversarial attacks.

This suggests that future defense should focus on higher-order features, such as sentence structure, coherence, and logical consistency, rather than just low-level character sequences.

5.5 Error Analysis

Our error analysis reveals that while LLMs are highly effective at style obfuscation, they are not invincible. We identified two primary reasons for false negatives where the detector failed to identify the adversarial text.

- **Model Stylistic Inertia**: Despite explicit instructions to mimic a specific human style, certain models—particularly LLaMA-3—exhibit what we term "stylistic inertia." This refers to the model's tendency to revert to its native generative bias. For instance, LLaMA-3 frequently overuses logical connectives such as "furthermore," "consequently," and "moreover". These stable transition patterns act as a secondary machine fingerprint. As Wang et al. (2025) suggest, these high-level discourse markers are often more resistant to style washing than low-level character n-grams, allowing the detector to occasionally flag even "Hard" imitation samples.

- **Stylistic Neutralization in Standardized Contexts**: We observed that detection performance significantly drops when the source text belongs to highly standardized domains, such as encyclopedic entries or medical reports. In these genres, human authors are already constrained by professional register and specific jargon, which naturally limits their personal stylistic variance (Stamatatos, 2009). When an LLM performs "style translation" on such texts, the resulting output is often indistinguishable from the human original because the target style itself is neutralized and lacks unique idiosyncratic fingerprints. This suggests that the current threat of LLM style imitation is most severe in formal, professional writing.

This finding indicates that future research in forensic linguistics should pivot toward **author identification in personalized, informal contexts**, such as social media posts or personal correspondence, where individual stylistic fingerprints are more pronounced and potentially harder for LLMs to replicate perfectly (Macko et al., 2024).

6 Conclusion

This paper builds a progressive adversarial evaluation framework and uses the HC3-Plus, MixSet, and MULTITuDE datasets to study the threat of large language models (LLMs) as "style translators" to authorship verification (AV). Unlike previous studies that only compare performance, we focus on a forensic perspective and show how stylistic evidence gradually fades under increasing adversarial strength.(Macko et al., 2023).

Our key findings are as follows:

- **Explicit Style Imitation**: High-performance models like GPT-4 can effectively weaken machine fingerprints. This causes detector performance to drop from nearly perfect (AUC 0.996) to close to random guessing (AUC 0.678) in hard imitation scenarios (Macko et al., 2024).

- **Attack Dimension Effects**: Semantic-level rewriting, such as GPT-4 paraphrasing, is the most destructive attack, reducing detection performance by approximately 17.6% on average. In contrast, physical-level attacks like homoglyphs have limited impact.

These results highlight an urgent point: as LLMs improve their instruction-following and style alignment, detection methods that rely only on surface-level statistics (character or word frequency) are becoming unreliable. Future AV and AI text detection research should focus on higher-order, cross-level stylistic features, including discourse structure, information organization, and logical coherence.

7 Limitations

Despite our systematic analysis, several limitations remain in this study:

1. **Limited coverage of attack methods**: Although we used the MULTITuDE adversarial subset to evaluate semantic, structural, and physical attacks (Macko et al., 2023), we did not cover all possible attack variants due to resource limits. Furthermore, most of our experiments focused on English texts. Cross-linguistic verification is still needed for languages with richer morphology.

2. **Black-box bias**: Some datasets, such as MixSet, do not disclose prompt details or specific decoding parameters. This makes it difficult to precisely measure the mathematical relationship between "imitation strength" and the detection escape rate.

3. **Model heterogeneity**: Due to API restrictions, we mainly tested GPT-4 and LLaMA-3. Future work should include more models, such as Claude, GPT5o, and explore advanced discriminative networks like contrastive learning or deep ensembles to check if these findings are universal.

4. **Limited professional domain coverage**: Our datasets primarily focus on general QA and news texts. Specialized domains like law, medicine, or creative writing may have stronger stylistic norms. The risks of "style hijacking" in these professional areas require further research.

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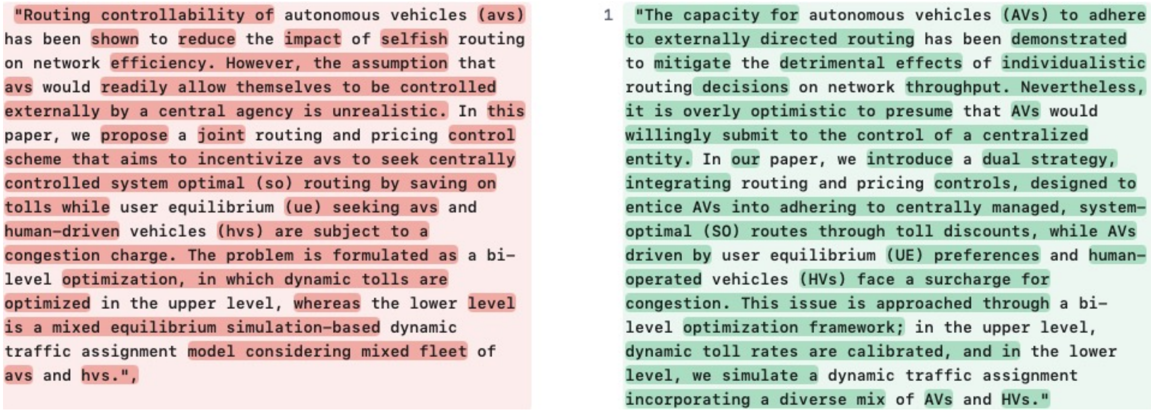
A Detailed Case Study of Style Obfuscation

To show how LLMs erase stylistic fingerprints, we look at a real example in Figure 3. This sample is from our **Pair 3 (Hard)** experiment. It shows how GPT-4 can change the writing style while keeping the meaning exactly the same (Zhang et al., 2024).

A.1 How the Text Changed

By looking at Figure 3, we can see a few ways the GPT-4 "Style Translator" hides machine traces:

- **Cleaning up inconsistencies**: Humans often use informal abbreviations (like "avs"). GPT-4 automatically changes these to standard academic forms like "AVs" (Bevendorff et al., 2024). This removes the small habits that detectors use to identify a writer (Stamatatos, 2009).


"Routing controllability of autonomous vehicles (avs) has been shown to reduce the impact of selfish routing on network efficiency. However, the assumption that avs would readily allow themselves to be controlled externally by a central agency is unrealistic. In this paper, we propose a joint routing and pricing control scheme that aims to incentivize avs to seek centrally controlled system optimal (so) routing by saving on tolls while user equilibrium (ue) seeking avs and human-driven vehicles (hvs) are subject to a congestion charge. The problem is formulated as a bi-level optimization, in which dynamic tolls are optimized in the upper level, whereas the lower level is a mixed equilibrium simulation-based dynamic traffic assignment model considering mixed fleet of avs and hvs.",

1 "The capacity for autonomous vehicles (AVs) to adhere to externally directed routing has been demonstrated to mitigate the detrimental effects of individualistic routing decisions on network throughput. Nevertheless, it is overly optimistic to presume that AVs would willingly submit to the control of a centralized entity. In our paper, we introduce a dual strategy, integrating routing and pricing controls, designed to entice AVs into adhering to centrally managed, system-optimal (SO) routes through toll discounts, while AVs driven by user equilibrium (UE) preferences and human-operated vehicles (HVs) face a surcharge for congestion. This issue is approached through a bi-level optimization framework; in the upper level, dynamic toll rates are calibrated, and in the lower level, we simulate a dynamic traffic assignment incorporating a diverse mix of AVs and HVs."

Figure 3: A side-by-side comparison of human-written text (left) and the GPT-4 polished version (right). The highlights show how the model changes words and tone to hide the author's identity (Zhang et al., 2024).

- **Using more formal words**: Simple phrases like "shown to reduce" are replaced with professional terms like "demonstrated to mitigate" (Zhang et al., 2024). This changes the character patterns (n-grams) that the detector is looking for (Stamatatos, 2009).

- **Expanding sentences**: The model turns short phrases into longer ones (Macko et al., 2024). For example, "selfish routing" becomes "individualistic routing decisions" (Zhang et al., 2024). This makes it much harder for a statistical detector to find the original "fingerprint" (Emmery et al., 2018).

This example proves that an LLM does not need to change the meaning of a sentence to fool a detector. By polishing the text and making it formal, the model successfully hides its identity (Zhang et al., 2024).